

Physics

Ans-1 - Convex lens

Ans-2: Convex lens of 10cm focal length has greater power

Ans-3: focal length is define as the distance between the centre of the lens on principal axis and its focal length point

$$v = -5 \text{ cm}$$

$$u = 10 \text{ cm}$$

Using lens formula

$$\frac{1}{f} = \frac{1}{u} - \frac{1}{v}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{(-5)}$$

$$\frac{1}{f} = \frac{3}{10}$$

$f = \frac{10}{3}$

~~f =~~

Ans-4: Using lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Hence v and u are the position of the image and object with respect to lens.

Magnification is the ratio of the size of the image and the size of the object

$$m = \frac{v}{u}$$

Hence image is inverted

$$-1 = \frac{v}{u}$$

$$v = -u$$

Given that image is formed at the distance from +50 from the lens. And since the image is real

$$u = +50 \quad \text{and} \quad v = -50$$

$$\frac{1}{f} = \frac{1}{50} - \frac{(-1)}{50}$$

$$\frac{1}{f} = \frac{2}{50} = \frac{1}{25}$$

$$f = 25$$

Power of the $P = \frac{1}{f}$

$$P = \frac{1}{25} \times 10^{-2} \text{ m}^{-1}$$

$$P = \frac{100}{25} = 4 \text{ m}^{-1}$$

$$R = 1 \text{ m}^{-1} = 10$$

$$S_0 \quad P = 40$$

$$P = 40$$